

# Formulation of Linear Programming Problems

## From Word Problems to Mathematical Model

Omkar Kashid

M.Sc. Mathematics

# Introduction

A Linear Programming Problem (LPP) is a mathematical model used to optimize a linear objective function subject to some constraints.

## Main Idea

We convert a real-life word problem into mathematical equations and inequalities.

The formulation of an LPP mainly consists of:

- Decision Variables
- Objective Function
- Constraints
- Non-negativity Restrictions

# Components of an LPP

## 1. Decision Variables

Unknown quantities that we need to determine.

$$x_1, x_2, \dots, x_n$$

## 2. Objective Function

The quantity which we want to maximize or minimize.

$$\text{Maximize/Minimize } Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

## 3. Constraints

Restrictions on the decision variables.

$$a_{11}x_1 + a_{12}x_2 \leq b_1$$

## 4. Non-negativity

$$x_1, x_2, \dots, x_n \geq 0$$

# Steps for Formulating a Word Problem

- 1 Read the problem carefully.
- 2 Identify the unknown quantities.
- 3 Define the decision variables.
- 4 Identify the objective.
- 5 Write the objective function.
- 6 Identify the available resources or restrictions.
- 7 Write the constraints.
- 8 Add non-negativity restrictions.

## Final LPP

Decision Variables + Objective Function + Constraints + Non-negativity Restrictions

# Important Words in Word Problems

| Word/Phrase      | Mathematical Form   |
|------------------|---------------------|
| At most          | $\leq$              |
| At least         | $\geq$              |
| Maximum          | Maximize            |
| Minimum          | Minimize            |
| Exactly          | $=$                 |
| Cannot exceed    | $\leq$              |
| Must be at least | $\geq$              |
| Available        | Resource constraint |
| Limited          | Resource constraint |

## Example 1: Maximization Problem

A company produces two products  $A$  and  $B$ .

Each unit of  $A$  gives a profit of Rs.40 and each unit of  $B$  gives a profit of Rs.60.

Let

$x$  = number of units of  $A$

$y$  = number of units of  $B$

The company wants to **maximize profit**.

Therefore,

$$\text{Maximize } Z = 40x + 60y$$

## Example 1: Constraints

Suppose each unit of  $A$  requires 2 hours of labour and each unit of  $B$  requires 3 hours.

Total labour available = 120 hours.

Therefore,

$$2x + 3y \leq 120$$

Suppose each unit of  $A$  requires 1 kg of material and each unit of  $B$  requires 2 kg.

Total material available = 80 kg.

Therefore,

$$x + 2y \leq 80$$

Hence the complete LPP is

$$\begin{aligned} &\text{Maximize } Z = 40x + 60y \\ &\text{subject to } 2x + 3y \leq 120, \\ &\quad x + 2y \leq 80, \\ &\quad x, y \geq 0. \end{aligned}$$

## Example 2: Minimization Problem

A person wants to prepare a diet using two foods  $A$  and  $B$ .  
Food  $A$  costs Rs.20 per unit and food  $B$  costs Rs.30 per unit.  
Let

$x$  = units of food  $A$

$y$  = units of food  $B$

The objective is to **minimize the total cost**.  
Therefore,

$$\text{Minimize } Z = 20x + 30y$$

## Example 2: Constraints

Suppose at least 40 units of nutrition are required.

Food  $A$  provides 2 units of nutrition and food  $B$  provides 4 units.

Therefore,

$$2x + 4y \geq 40$$

Suppose another nutritional requirement is

$$3x + 2y \geq 30$$

Hence,

$$\begin{aligned} &\text{Minimize } Z = 20x + 30y \\ &\text{subject to } 2x + 4y \geq 40, \\ &\quad 3x + 2y \geq 30, \\ &\quad x, y \geq 0. \end{aligned}$$

## Example 3: “At Most”

A factory produces products  $A$  and  $B$ .

The total number of products produced cannot exceed 50.

Let

$$x = \text{number of } A, \quad y = \text{number of } B.$$

The phrase

**“cannot exceed 50”**

means

$$x + y \leq 50$$

Thus, “at most” and “cannot exceed” generally give a  $\leq$  constraint.

## Example 4: “At Least”

A company must produce at least 100 units of products.  
Let

$x$  = number of product A

$y$  = number of product B

The phrase “at least 100” means

$$x + y \geq 100$$

Thus,

**At least**  $\Rightarrow \geq$

## Example 5: Exactly

A company needs exactly 500 kg of raw material.

If product  $A$  requires 2 kg and product  $B$  requires 3 kg, then

$$2x + 3y = 500$$

Therefore,

**Exactly**  $\Rightarrow =$

This is an equality constraint.

## Example 6: Production Problem

A furniture company produces tables and chairs.

Let,

$x$  = number of tables

$y$  = number of chairs

Each table requires 4 hours of labour and each chair requires 2 hours.

Available labour = 100 hours.

Therefore,

$$4x + 2y \leq 100$$

If profit per table is Rs.50 and per chair is Rs.30,

$$\text{Maximize } Z = 50x + 30y$$

with

$$x, y \geq 0.$$

## Example 7: Minimum Cost Problem

A transport company uses two types of vehicles.

Let,

$x$  = number of type 1 vehicles

$y$  = number of type 2 vehicles

Cost per vehicle:

$$C_1 = \text{Rs.}500, \quad C_2 = \text{Rs.}700$$

The company wants to satisfy the transportation requirement at minimum cost.

Therefore,

$$\text{Minimize } Z = 500x + 700y$$

The transportation requirements are written as constraints.

# General Form of an LPP

For  $n$  decision variables:

Maximize/Minimize

$$Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

subject to

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq, \geq, = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq, \geq, = b_2$$

$$\vdots$$

$$x_1, x_2, \dots, x_n \geq 0.$$

# Conclusion

The formulation of an LPP converts a real-life word problem into a mathematical model.

The main steps are:

- 1 Define decision variables.
- 2 Formulate the objective function.
- 3 Formulate constraints.
- 4 Add non-negativity restrictions.

## Key Point

Correct formulation is the first and most important step before solving an LPP by Simplex, Graphical Method, Big-M, Two-Phase Method, etc.

# Thank You